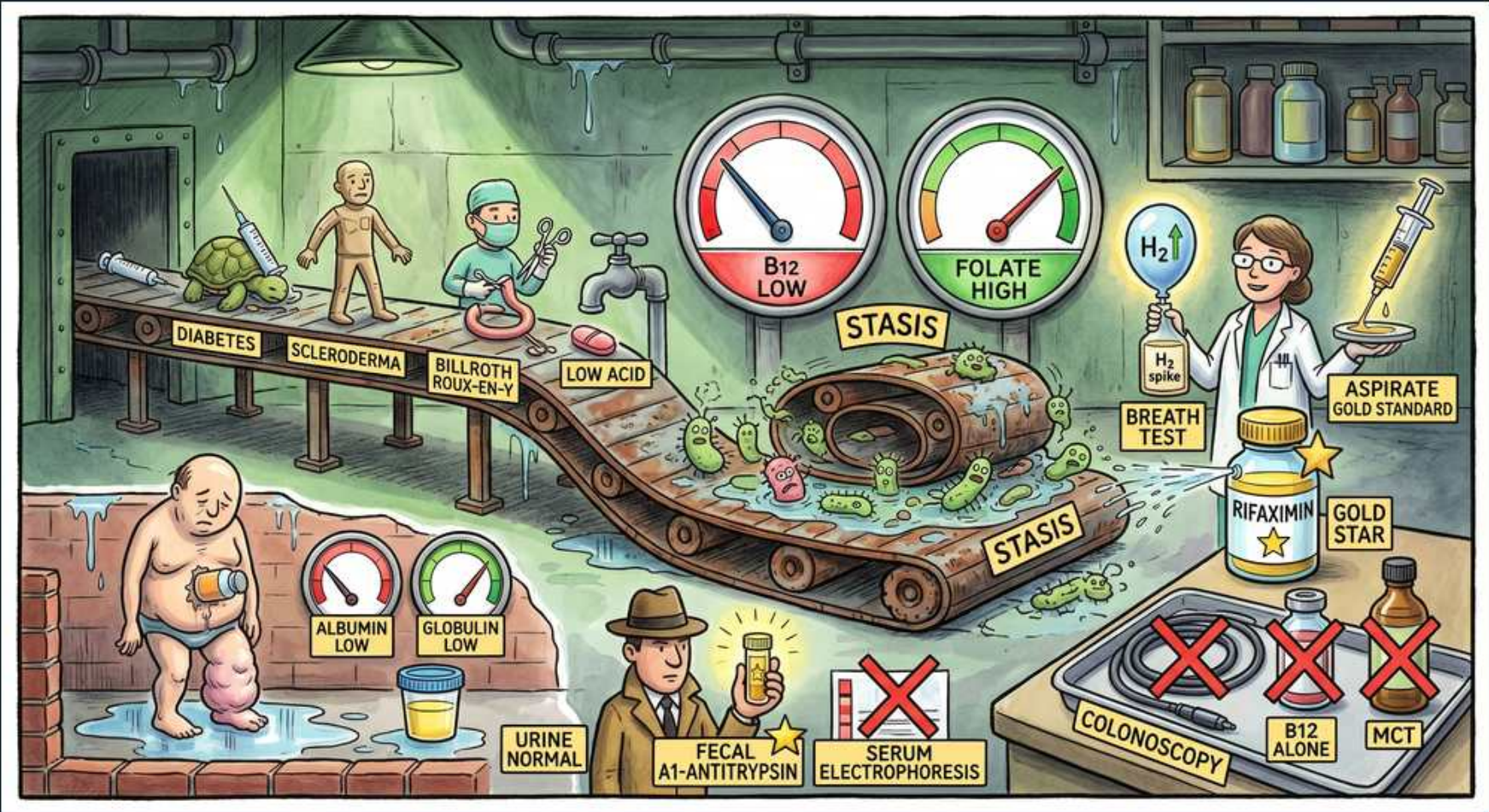


Malabsorption — SIBO & Protein-Losing Enteropathy



1. Diabetes (stasis)

WHAT YOU SEE

A turtle and 'DIABETES' tag at the start of the conveyor belt, left.

WHAT IT ENCODES

SIBO arises from intestinal STASIS. Diabetes causes autonomic neuropathy with slowed gut motility (gastroparesis/dysmotility), letting bacteria overgrow. Slow transit is the unifying mechanism (turtle = slow).

2. Scleroderma (stasis)

WHAT YOU SEE

A stiff figure labeled 'SCLERODERMA' on the conveyor, left-center.

WHAT IT ENCODES

Scleroderma causes smooth-muscle atrophy/fibrosis of the small bowel wall -> dysmotility and stasis -> SIBO. A classic GI complication of systemic sclerosis presenting with bloating, diarrhea, weight loss.

3. Billroth/Roux-en-Y blind loop

WHAT YOU SEE

A surgeon figure 'BILLROTH ROUX-EN-Y' on the conveyor, center-left.

WHAT IT ENCODES

Surgical blind/afferent loops (Billroth II, Roux-en-Y) and strictures/diverticula create stagnant segments where bacteria accumulate ('blind loop syndrome'). Anatomic stasis is a major SIBO cause.

4. Low gastric acid

WHAT YOU SEE

A faucet/tap labeled 'LOW ACID', center.

WHAT IT ENCODES

Low gastric acid (chronic PPI use, atrophic gastritis, prior surgery) removes the acid barrier that normally limits bacterial colonization, promoting SIBO. Strictures and impaired immunity also contribute.

5. B12 LOW (gold-star pattern)

WHAT YOU SEE

A red gauge 'B12 LOW' over the conveyor, center.

WHAT IT ENCODES

Signature of SIBO: B12 is LOW because the overgrowing bacteria CONSUME dietary/intrinsic-factor-bound B12 before the terminal ileum can absorb it. Low B12 + high folate is the classic discriminating pair on the boards.

6. Folate HIGH (gold-star pattern)

WHAT YOU SEE

A green gauge 'FOLATE HIGH' next to the B12 gauge, center.

WHAT IT ENCODES

Folate is HIGH in SIBO because gut bacteria SYNTHESIZE folate. The combination LOW B12 + HIGH folate strongly suggests SIBO and distinguishes it from other malabsorption (where folate is usually low). Expecting low folate is the trap.

BOARD TRAP

"Folate is low in SIBO" → wrong; bacteria synthesize folate, so folate is typically HIGH while B12 is low.

7. Breath test (Dx)

WHAT YOU SEE

A clinician holding an 'H2 spike / BREATH TEST' flask, right-center.

WHAT IT ENCODES

Diagnose SIBO with a hydrogen/methane breath test (glucose or lactulose) showing an early rise in breath H2/CH4. Non-invasive and first-line in practice, though imperfect.

8. Jejunal aspirate = gold standard

WHAT YOU SEE

A syringe drawing fluid labeled 'ASPIRATE GOLD STANDARD', upper-right.

WHAT IT ENCODES

Jejunal (small-bowel) aspirate culture is the GOLD STANDARD for SIBO: >=10^3-10^5 CFU/mL is diagnostic. Invasive and less practical than breath testing, but the reference standard.

9. Rifaximin (gold star)

WHAT YOU SEE

A 'RIFAXIMIN' bottle with gold star and 'GOLD STAR' tag, right-center.

WHAT IT ENCODES

Treat SIBO by eradicating the overgrowth with rifaximin (gold star), a non-absorbed antibiotic; alternatives include amoxicillin-clavulanate or metronidazole. ALSO correct the underlying cause (motility, anatomy) and replete B12. Antibiotic — not vitamins — is the fix.

10. Colonoscopy / B12 alone / MCT (X)

WHAT YOU SEE

Three crossed-out items on a stove: 'COLONOSCOPY', 'B12 ALONE', 'MCT', lower-right.

WHAT IT ENCODES

Distractors: colonoscopy does NOT diagnose SIBO (it's a small-bowel/breath-test problem); giving B12 ALONE ignores the cause (overgrowth recurs); MCT oil is not the SIBO fix. Treat the bacteria (rifaximin) and the stasis.

BOARD TRAP

"Diagnose SIBO with colonoscopy / treat with B12 alone / fix with MCT" → all wrong; diagnose with breath test or jejunal aspirate and treat with rifaximin plus correcting the cause.

11. PLE: low albumin AND globulin

WHAT YOU SEE

A swollen child with gauges 'ALBUMIN LOW' and 'GLOBULIN LOW', lower-left.

WHAT IT ENCODES

Protein-losing enteropathy: gut loses protein non-selectively, so BOTH albumin AND globulin (immunoglobulins) fall together — unlike nephrotic syndrome (albumin lost, urine abnormal) or cirrhosis. Causes edema. Look for Menetrier, lymphangiectasia, IBD.

12. Urine normal

WHAT YOU SEE

A 'URINE NORMAL' label near the PLE child, lower-left.

WHAT IT ENCODES

In PLE the URINE is NORMAL (no proteinuria), which separates it from nephrotic syndrome where protein is lost renally. Combined low albumin + low globulin + normal urine = think gut protein loss.

13. Fecal alpha-1-antitrypsin (gold star)

WHAT YOU SEE

A detective holding a stool jar 'FECAL A1-ANTITRYPSIN' with gold star, lower-center.

WHAT IT ENCODES

Diagnose PLE with FECAL alpha-1-antitrypsin clearance (gold star) — A1AT is a serum protein not absorbed or secreted by the gut, so its appearance in stool proves enteric protein loss. Serum protein electrophoresis is the wrong test here.

14. Serum electrophoresis (X)

WHAT YOU SEE

A 'SERUM ELECTROPHORESIS' panel with a red X, lower-center.

WHAT IT ENCODES

Serum protein electrophoresis does NOT diagnose protein-losing enteropathy — it only shows the low proteins, not where they go. The confirmatory test is fecal alpha-1-antitrypsin. SPEP is a distractor.

BOARD TRAP

"Use serum protein electrophoresis to diagnose PLE" → wrong; SPEP only quantifies the low proteins. Confirm enteric loss with fecal alpha-1-antitrypsin.